Imtiaz Ahammad

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Professional Summary

AI Engineer with 2+ years of production experience building computer vision and LLM-powered systems. Specialized in real-time ANPR and traffic analytics pipelines, multi-camera video processing on edge devices (NVIDIA Jetson, Raspberry Pi), and RAG-based LLM applications. Currently leading a 4-person AI team deploying intelligent toll infrastructure for national smart-city projects in Bangladesh. Experienced with the full ML lifecycle: data annotation, model training, optimization (TensorRT/ONNX), deployment, and field validation.

Work Experience

Sept 2025 – Present

AI Engineer (Team Lead), **Regnum Resource Ltd**, *Dhaka, Bangladesh.*

Leading a 4-person AI team building production computer vision systems for Bangladesh's national toll and traffic infrastructure.

1. Architected and deployed an end-to-end intelligent toll management system covering ANPR, RFID automation, barrier control, and vehicle analytics — now processing high-volume vehicle streams across multiple toll points using YOLO-based detectors and Transformer-based vision models running on NVIDIA DeepStream-powered edge devices.
2. Built a real-time Traffic Flow Analysis System ingesting multi-camera RTSP streams (8+ feeds) using RFDETR and ByteTrack for vehicle classification and density estimation; engineered specific hardware requirements for high-concurrency Computer Vision and LLM architectures, integrated automated data pipelines into Oracle 19c for large-scale analytics, reducing manual reporting time by approximately 50%.
3. Integrated production AI modules with existing toll collection software, IoT sensors, and government e-governance infrastructure; managed annotation pipelines in Roboflow and maintained Git-based deployment workflows ensuring compliance with national smart-city standards.
4. Mentored three interns through the full ML workflow — dataset creation, model training and evaluation, deployment, and field-level validation of computer vision systems in live toll environments.

March 2024 – August 2025

Machine Learning Engineer, **ANTT Robotics**, *Dhaka, Bangladesh.*

1. Fine-tuned LLaMA and GPT models to build **ANTT AI Bot**, a hardware-specific code-generation assistant for Arduino, ESP32, and Raspberry Pi — reducing student onboarding time by ∼40% and achieving an ∼85% pass rate on functional hardware validation tests.
2. Developed **Predictive Maintenance** pipelines using ARIMA, Prophet, and XGBoost with automated weekly model retraining; deployed production inference APIs on AWS EC2 via FastAPI, achieving sub-200ms response latency under concurrent load.
3. Designed a **Real-time ANPR and Dynamic Toll Collection** system using YOLOv8 and PaddleOCR to extract number plates, classify vehicle types, and compute tolls dynamically; integrated with HD camera pipelines and a PyQt5 operator dashboard, reaching >90% plate-read accuracy under varied lighting conditions.
4. Engineered a **Driver Behavior and Safety Monitoring** system using MediaPipe Face Mesh and Dlib to compute MAR, EAR, and Head Pose Estimation in real time, triggering fatigue alerts with <500ms detection latency.
5. Developed a **RAG-based Recommendation Chatbot** for retail stores combining ChromaDB vector search with GPT-4o, served via FastAPI with a ReactJS frontend — deployed and used by 3 pilot retail partners.

Technical Skills

- **CV & Video Analytics:** YOLO (v5–v26, custom training, TensorRT export), RFDETR, Transformer-based detectors, OpenCV, MediaPipe, Albumentations, ByteTrack, DeepSORT, SAM, RTSP, multi-camera pipelines, NVIDIA DeepStream
- **OCR & Perception:** PaddleOCR, EasyOCR, Tesseract, Multimodal OCR, Face Landmark Detection, Head Pose Estimation, Keypoint & Human Pose Tracking
- **LLM & RAG Systems:** LangChain, LangGraph, LlamaIndex, RAG pipelines (dense + sparse hybrid), Prompt Engineering, Fine-tuning (LLaMA, GPT), Vision-Language Models, RAGAS evaluation, FastMCP / MCP integration
- **Model Optimization & MLOps:** TensorRT, ONNX Runtime, FP16/INT8 quantization, Pruning, Docker, FastAPI microservices, AWS EC2, Git, Roboflow annotation pipelines
- **Edge & Embedded Deployment:** NVIDIA Jetson (Nano / Orin), Raspberry Pi, ONNX export, Pyserial (IoT/sensor integration), PyQt5
- **ML Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, ONNX Runtime, PaddlePaddle
- **Databases & Vector Search:** PostgreSQL (pgvector), Oracle 19c / 26ai, MySQL, FAISS, ChromaDB, Redis (semantic cache)
- **Time-Series & Classical ML:** ARIMA, Prophet, XGBoost, Gradient Boosting, EDA, Feature Engineering

Selected Personal Projects

1. **Zero-Trust Multi-Agent Knowledge Mesh RAG with MCP** [**Project Link**]
   - Engineered an enterprise-grade RAG pipeline with document-level RBAC enforced via custom JWT middleware, injecting dynamic department and clearance metadata filters into pgvector queries.
   - Orchestrated a data mesh across distributed mock enterprise endpoints (Confluence, Jira, Slack) using Anthropic's Model Context Protocol (MCP) with FastMCP servers communicating over WebSockets/SSE.
   - Built a hybrid search engine in PostgreSQL combining BGE-Large dense embeddings (pgvector) with BM25 sparse keyword search; merged results via Reciprocal Rank Fusion (RRF) and FlashRank cross-encoder reranking.
   - Reduced downstream vLLM inference calls by building a Redis semantic cache (0.92 cosine similarity threshold); validated full pipeline quality using automated RAGAS metrics.
   - **Stack:** PostgreSQL (pgvector), Redis, FastMCP (MCP), FastAPI, vLLM, FlashRank, RAGAS, Docker, HyDE
2. **Local Incident Resolution Graph for AIOps** [**Project Link**]
   - Architected a privacy-first AIOps agent using Oracle 26ai hybrid vector search and HNSW indexing to

autonomously map PagerDuty alerts to historical runbooks, eliminating cloud API data-leakage risk for sensitive financial logs.
  - Chose Qwen 2.5 Coder 14B (quantized, via Ollama) over GPT-4 to enable fully local LLM orchestration on consumer hardware — achieving 1.7s warm response latency with no external data transmission.
  - Integrated SentenceTransformers embeddings with structured Oracle relational joins for deterministic incident synthesis over custom-seeded historical telemetry.
  - **Stack:** Oracle 26ai, LangChain, Qwen 2.5 Coder (Ollama), SentenceTransformers, FastAPI, Streamlit, Pytest
3. **GiftSpark: AI-Powered Gift Recommendation Engine** [**Project Link**]
  - Architected a multi-stage RAG pipeline combining rule-based filters with ChromaDB vector similarity search to surface personalized gift recommendations, improving retrieval precision over keyword search by 3×.
  - **Stack:** GPT-4o, ChromaDB, OpenAI Embeddings, LangChain, FastAPI
4. **Face Mesh and Behavioral Analysis System** [**Project Link**]
  - Built a real-time 3D facial landmark tracking system using MediaPipe Face Mesh and OpenCV, enabling downstream behavioral analysis features (blink rate, gaze estimation, expression tracking).
  - **Stack:** MediaPipe, OpenCV, PyQt5, cvzone

Publications

Conference Paper

- **Imtiaz Ahammad**, Suranjeet Chowdhury Avik, Abdullahi Chowdhury, Moriwam, MD Abdullah Al Maruf, and Ranesh Naha. **"AI-Powered Resilience: Addressing the Mental Health Impact of Mass Layoffs in the Digital Age."** arXiv preprint arXiv:2405.04610 (2024). **[Published at ICETITE 2024] [IEEE Link]**

Academic Thesis (Under Review)

- **Imtiaz Ahammad**, Moriwam, Md Sabbir Ahmed. **"Leveraging PostHoc XAI on Blackbox Models: A Holistic Approach to Model Interpretability and Transparent Veracity Assessment with the Largest Verified Dataset."** **[Under Review — International Journal of Information Management Data Insights]**

Awards & Achievements

Oct 2023 **1st Place — Sustainable Business Idea Competition 2023**, World School of Business

Proposed **InCareer**, an AI-powered soft-skill EdTech and RFP-matching job portal for fresh graduates. **[Event Link]**

Education

2020 – 2024 **World University of Bangladesh**, Dhaka-1230, Bangladesh

B.Sc. in Computer Science and Engineering

References

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Nazib Ahmed

Co-founder & CTO, ANTT Robotics

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